Erin Payne

erin.payne.01@gmail.com | linkedin.com/in/erin-payne3141 | 604-741-7787

Education

BASc, Mechatronics and Robotics Engineering, Queen's University, Kingston ON — Sept. 2024 – May 2028

Work Experience

Mechatronics Design Intern – Great Circle Solar Management Corporation — Apr. 2026 – Aug. 2026

- Designed and optimized CAD models and a distributed control system for networked devices, supporting scalable deployment of solar panel snow removal across utility-scale solar farms.
- Conducted on-site testing, troubleshooting, and iterative design improvements to enhance mechanical performance, reliability, and system robustness in harsh outdoor environments.
- Optimized embedded control software and communication protocols by redesigning the RS485 network architecture and refactoring legacy firmware into a modular, maintainable system.

Extra-Curricular Experience

Co-Captain - Queen's Aerospace Design Team — May 2026 – Present

- Leading a multidisciplinary team of 160+ students in the design, fabrication, and testing of a carbon fiber UAV for the SAE Aero Design Competition, overseeing technical development, project execution, and systems integration.
- Manage team operations, including budgeting, sponsorships, recruitment, project timelines, and competition logistics, while coordinating cross-functional engineering teams to deliver a competitive aircraft on schedule.

Avionics Manager - Queen's Aerospace Design Team — May 2025 – May 2026

- Lead a team of 10+ to develop flight-ready avionics systems for two fixed-wing VTOLs and a quadcopter, enabling urban-firefighting missions including fire reconnaissance, equipment deployment, and indoor autonomous fire mitigation.
- Conducted systems testing, including full flight performance reviews using sensor data, and flight logs.
- Oversaw component selection, procurement, and cost tracking to stay within project budget and timeline.
- Coordinated with structures, aerodynamics, software, and propulsion systems teams to align systems requirements and optimized performance for all aircraft.

Electrical and Power Systems Team General Member - Queen's Aerospace Design Team — Oct. 2024 – Apr. 2025

- Contributed to the design of a carbon fiber fixed wing and quadcopter drone for wildfire hotspot detection and water transport.
- Focused on electronics and power systems, helped select optimal components and integrated GPS modules, cameras, and QGroundControl compatible components.
- Assisted in power distribution hardware design, battery configuration and soldering, learned the basics of power distribution board schematics using LTSpice and Altium, helped electrical team integrate a 12S 2P battery with quick-swapping capabilities.

Royal Canadian Marine Search and Rescue Volunteer - RCM-SAR Station #14 — Apr. 2023 – Aug. 2024

- Youth volunteer trained in helm, navigation, logbook, and radio communication roles, developed docking, navigation, shore searches, anchoring, and communication skills.
- Participated in monthly station meetings covering task debriefs, first-aid updates, safety and training reports, and supported monthly boat maintenance on the Falkins Class Type-1 fast response vessel.
- Participated in technical classroom sessions on marine-specific first aid, chart plotting, meteorology and navigation.

Projects

Avionics Integration and Controls Tuning of UAV — Summer 2025

- Tuned and calibrated PID parameters in QGroundControl to improve flight stability and control responsiveness.
- Conducted structured test flights, analyzed logs, and isolated issues such as rate-gain imbalance, interference and sensor noise.
- Refined rate and attitude loops through controlled parameter sweeps, reducing oscillations and achieving smoother, more predictable flight behaviour, implemented test rig to improve safety and reduce risk of airframe damage while testing.

Autonomous Mars Sample Return Rover (MREN 203 Group Project) — Spring 2026

- Developed an Autonomous Rover with a robust, roller intake system designed for reliable operation in harsh environments.
- Implemented autonomy stack using LiDAR SLAM, wheel odometry, path planning for navigation and obstacle avoidance.
- Designed and tuned closed-loop control and perception systems to enable stable autonomous navigation and sample retrieval.

Technical Skills

Electrical/Software Px4, QGroundControl, LTSpice, Arduino, soldering, C/C++, Python, Git, VHDL, Quartus Prime, Julia
Mechanical: SolidWorks, OnShape, FreeCAD, drafting, 3D Printing
Platforms & Tools: MS Excel, MS Word, MS PowerPoint